QUANT X CLUB · ALPHA X RESEARCH PROGRAM

The Three Edges

A beginner's masterclass in selling options premium

01 Variance Risk Premium

02 IV-Crush Capture

03 GEX Regime Switch

From "why selling options works" → to building & backtesting on Options Alpha

SIC Capital Pte. Ltd.

HOW THIS DECK WORKS

Three edges, taught easiest-first

We reorder by difficulty, not by importance. Each edge builds on the one before it.

1 **Variance Risk Premium** — Tier B · OA

The bedrock: WHY selling premium pays at all. The insurance-company mindset.

2 **IV-Crush Capture** — Tier A/B · OA

A specific WHEN: the morning fear premium deflating on quiet days. Your own H1a finding.

3 **GEX Regime Switch** — Tier A · OA

The advanced one: reading dealer hedging to know when to sell vs stand aside.

PART 0

Foundations

The five ideas every edge below depends on.

FOUNDATIONS · 1 OF 4

An option premium is the price of a promise

- An option is a contract. The buyer pays money today for the right to buy or sell later at a fixed price.
- That money is the premium — paid up front, kept by the seller no matter what happens next.
- The BUYER wants protection or a lottery ticket. They pay the premium.
- **The SELLER collects the premium and takes on the obligation. That's the seat we sit in for all three edges tonight.**

🏡

Think: home insurance

- You pay a premium every year for fire cover.
- The insurer keeps it whether or not your house burns down.

Why sell? You become the insurance company

Insurers stay in business because they charge a little MORE than the claims they expect to pay. Option sellers do the same thing.

💰

Collect premium

Money in your pocket the moment you sell the option.

⏳

Wait & bleed time

Every day that passes, the option loses value. Time decay works for you.

🛡️

Manage the tail

Occasionally a 'fire' happens. Your whole job is surviving those.

The whole game:

The three numbers you need tonight

DTE

Days to Expiration

How long until the promise expires. Fewer days = faster time decay in your favour, but less room for error.

0DTE = expires today

Δ

Delta

Read it as the rough chance the option finishes in-the-money. A 0.20 delta short option ≈ 20% chance it hurts you.

0.20 ≈ 20% ITM odds

IV

Implied Volatility

The market's fear gauge — how big a move buyers are pricing in. High IV = fat premiums to sell.

High IV = rich premium

Structure vs Signal — the idea everything hangs on

THE STRUCTURE

What you trade

- The option legs: e.g. a short put spread at 20-delta, 7 DTE.
- The exits: profit target, stop loss, time-based close.

THE SIGNAL

WHEN you trade

- The gate that says 'today qualifies': high IV Rank, positive GEX, a rich RV/IV gap.
- This is where the real edge usually lives.

★ THE RULE

PART 1

Variance Risk Premium

Get paid because fear is systematically overpriced.

You charge for fear. You pay for what actually happens.

IMPLIED VOL = the premium you CHARGE

The market's guess of how wild the future will be. Buyers pay up for protection, so this guess runs high.

REALIZED VOL = the claims you PAY

How wild the market ACTUALLY turned out to be. On average, calmer than the fear priced in.

IV – RV

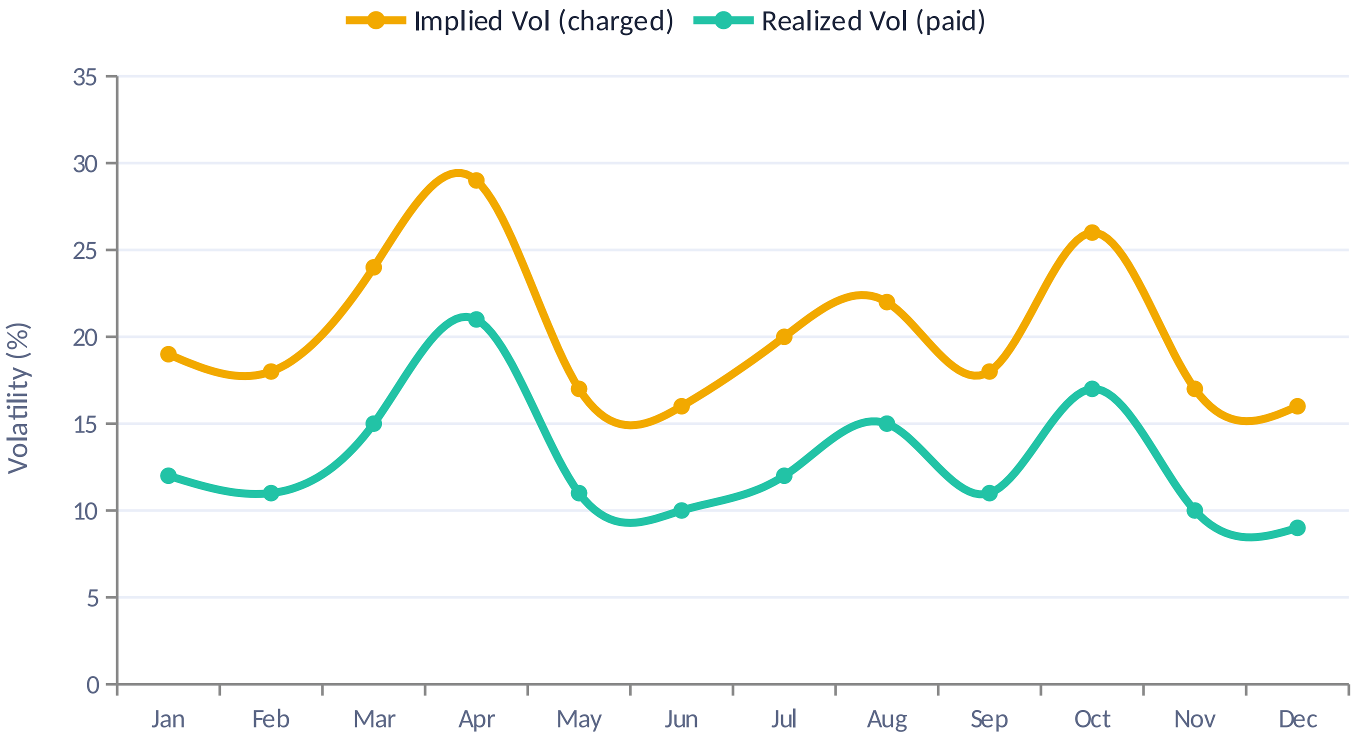
= *the variance risk premium*

Sell the overpriced fear, pay out the smaller reality, keep the difference — over and over.

VRP · THE RV/IV GATE

The gate: only sell when the premium is EXTRA fat

The gap exists most of the time — but it's fattest when fear spikes above what reality delivers. The gate waits for those days.



Read the chart

- Gold (IV) sits ABOVE teal (RV) almost every month.
- That persistent gap is the premium you harvest.
- **The gate fires when the gap is widest — Mar, Apr, Oct here.**

⚠️ Warning: the widest gaps often come right before a storm. The gate helps, but never removes tail risk.

VRP · WORKED EXAMPLE

Selling a put spread on a calm week

The trade

Underlying	**SPY at $500**
Structure	**Sell 7-DTE put spread**
Short leg	**$485 put (20-delta)**
Long leg	**$480 put (protection)**
Premium collected	**$1.10 ($110)**
Max loss	**$3.90 ($390)**
Gate check	**IV Rank 68 ✓ qualifies**

If SPY stays above $485 (the calm base case)

$110 premium. The options expire worthless — the 'fire' never happened, and you kept the insurance money.

◆ LIVE DEMO

Build this exact put spread live: enter the two legs and show the payoff diagram. Point to the flat green profit zone above $485 and the capped red loss below $480 — students SEE that risk is defined before entry.

This isn't folklore — it's decades of research

Bakshi & Kapadia (2003)

Delta-Hedged Gains and the Negative Market Volatility Risk Premium

Showed option sellers earn a premium even after hedging away direction — the vol premium is real and negative for buyers.

Carr & Wu (2009)

Variance Risk Premiums

The landmark study measuring IV minus RV across indices. Confirms IV systematically exceeds realized variance.

Black & Scholes (1973)

The Pricing of Options

The foundation — gives us the delta ≈ probability-of-ITM intuition we used for strike selection.

What OA can and can't do here

✓ OA does this natively

- Short-DTE put spread structure
- Strike selection by delta (e.g. 20-delta short)
- Sweep delta × spread width × DTE
- Gate on IV Rank or VIX level

✗ OA can't do this

- No realized-vol input series
- No 'IV minus trailing RV' comparison
- The true RV/IV gate can't be expressed
- Closest proxy: IV Rank (IV vs its OWN past, not vs realized)

◆ LIVE DEMO

Pull up Barchart's Implied vs Historical Volatility chart for SPY. Show the two lines and the gap — THIS is the signal OA can't see, which is why VRP lives on R2 / Options Omega.

The honest verdict

OA validates the STRUCTURE with an IV-Rank proxy gate. The real RV/IV edge is found and owned on R2. This is a Tier B strategy: brute-force on R2, sanity-check the structure on OA.

When the insurance company blows up

The catch: you win small and often, then occasionally lose big. One uncovered 'fire' can erase months of premium. This is the exact risk shape all three edges share.

1

Always define your loss

Use spreads, not naked options. The long leg caps the fire damage.

2

Size so no day can ruin you

1–2% risk per trade. A single loss must never threaten the account.

3

Respect the tail

Trend filters and event-skips (e.g. avoid FOMC) keep you out of the worst days.

PART 2

IV-Crush Capture

The morning fear premium deflates on quiet days.

IV-CRUSH · THE CORE IDEA

A balloon of fear, deflating through the day

- At the open, a 0DTE option is priced for a FULL day of possible chaos. Fear is maximal.
- As quiet hours pass with no big move, that fear premium leaks out — like air from a balloon.
- By the close, an option that expires in minutes has almost no 'anything could happen' value left.
- **A premium seller who sold in the morning buys back cheap (or lets it expire) — pocketing the deflation.**

9:35 AM

IV ≈ 17% — full balloon, priced for chaos

3:30 PM

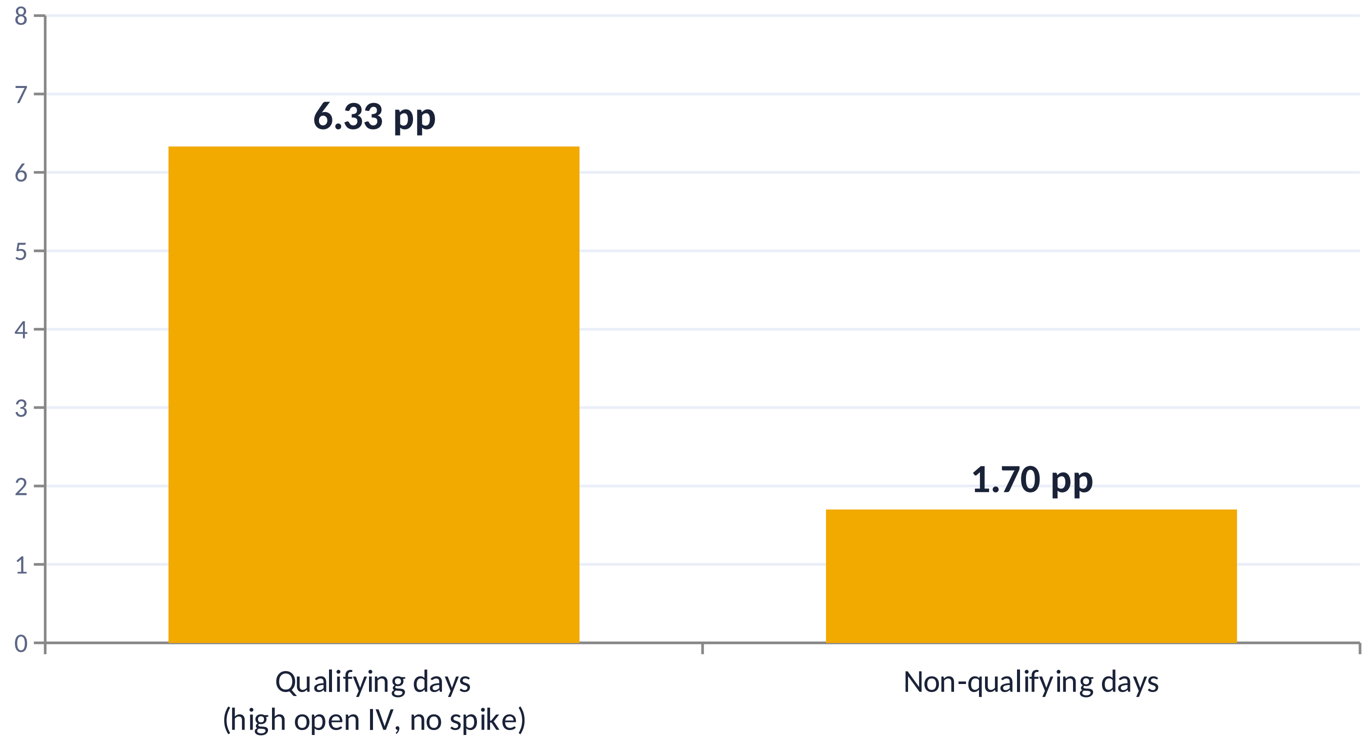
IV ≈ 9% — deflated, the day is nearly over

−47% IV crush on a quiet day

IV-CRUSH · YOUR OWN PROOF

You already validated this — the H1a result

From your EDA notebook: on qualifying quiet, high-IV days the crush is ~3.7× larger than on other days.

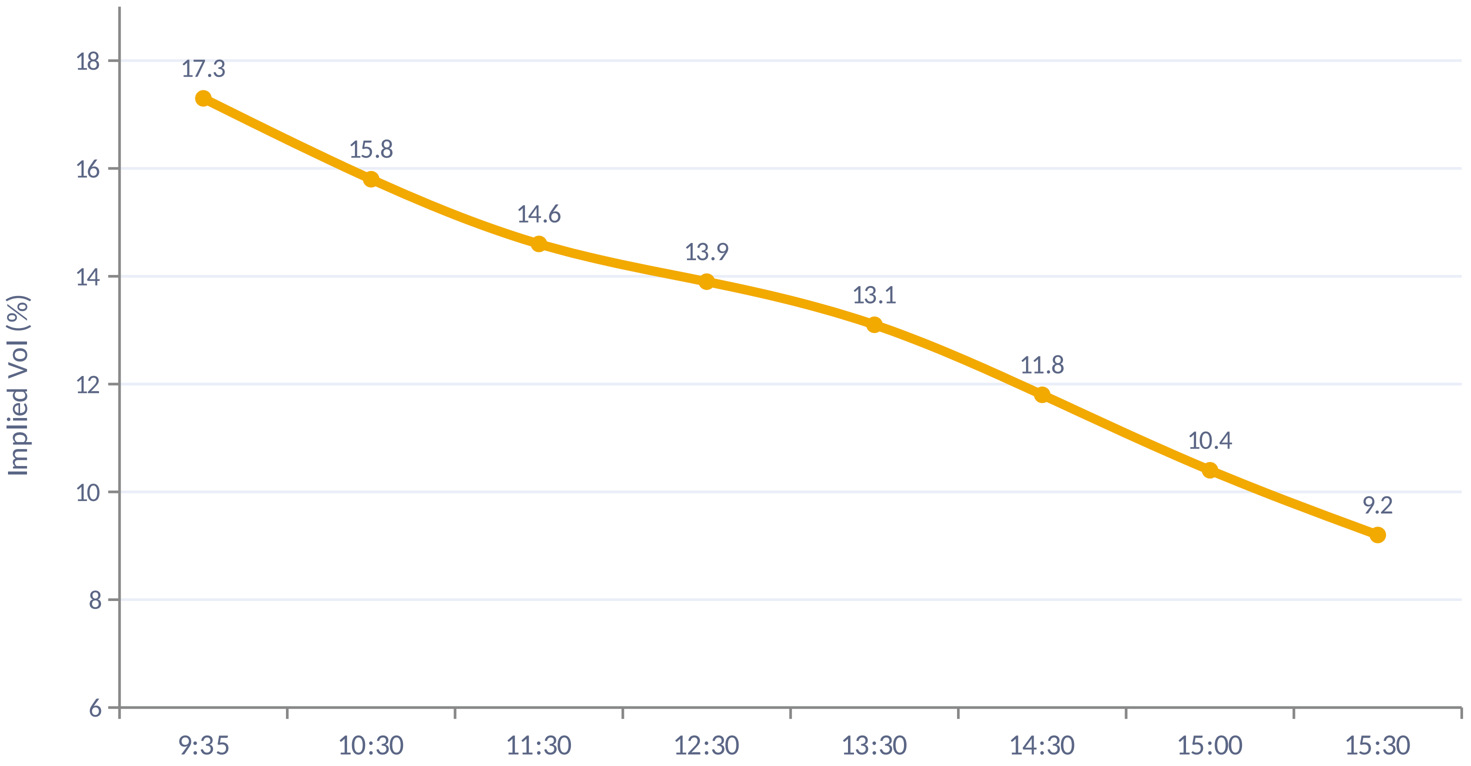


Why this matters tonight

- This edge isn't borrowed — YOUR team measured it on real R2 data.
- That filter is exactly what we'll try to reproduce on Options Alpha.

IV-CRUSH · WHAT IT LOOKS LIKE

One quiet Monday, minute by minute



The shape to know

- A steady downhill slide all day.
- **The crush ACCELERATES in the final 90 minutes — theta bites hardest near expiry.**
- Enter mid-morning, close before the bell to dodge settlement risk.

IV-CRUSH · THE EVIDENCE

The 0DTE research the pros are reading

Brogaard, Han & Won (2024)

Does 0DTE options trading increase volatility? Maps how 0DTE flow and dealer hedging move the underlying.

Dim, Eraker & Vilkov (2024)

0DTEs: Trading, Gamma Risk and Volatility Propagation — the mechanics of same-day option risk.

Beckmeyer, Branger & Gayda (2023)

Retail traders love 0DTE options… but should they? Sobering look at who actually profits.

Heston, Korajczyk & Sadka (2010)

Intraday return seasonality — the academic basis for time-of-day entry timing.

IV-CRUSH · BUILD IT ON OPTIONS ALPHA

Mostly native — with two proxy gates

✓ Native structure

- Time-of-day entry (e.g. 10:00)
- 0DTE ATM / near-ATM structure
- Profit-target % and close-before-close exits
- Sweep: entry time × structure × profit target

~ Signal via proxy

- “High open IV” → gate on IV Rank
- “Quiet day” → 'trading range < X% since prev close' filter
- **These are close cousins of your H1a filter — expect some slippage vs the notebook result.**

⚠️ The caveat that matters most

OA backtests the P&L of the STRUCTURE, not the crush itself. A crush can be real and the trade still lose if price runs through your strikes. “Crush confirmed” ≠ “strategy profitable.” Never let students conflate the two.

PART 3

GEX Regime Switch

Read dealer hedging to know when to sell — and when to hide.

GEX · THE CORE IDEA

Dealers: shock absorbers, or no suspension at all

🟢 POSITIVE GEX

Dealers are shock absorbers

- Their hedging pushes AGAINST moves.
- Price gets 'pinned' — calm, range-bound, mean-reverting.
- **→ Great weather for selling premium.**

🔴 NEGATIVE GEX

Dealers are a car with no suspension

- Their hedging AMPLIFIES moves.
- Price trends hard, gaps, accelerates — every bump gets bigger.
- **→ Dangerous for sellers. Stand aside.**

GEX · THE ONE NEW CONCEPT

Why dealer hedging pins or accelerates price

Taught slowly — this is the only genuinely new machinery tonight.

Dealers sell options to the public

Every 0DTE contract retail buys, a dealer is on the other side — and must hedge to stay neutral.

Hedging means buying/selling the underlying

To stay neutral as price moves, dealers trade SPY/SPX shares or futures constantly.

The DIRECTION of that hedging is the whole story

Positive gamma → they buy dips & sell rips (calming). Negative gamma → they sell dips & buy rips (amplifying).

GEX · SAME TRADE, OPPOSITE DAYS

One structure, two regimes

- You sell an iron condor at 9:45.
- Price drifts, dealers pin it near the open level.
- IV crushes, your condor decays.
- **Close at 50% profit. The pin did the work.**

- Same condor, same time.
- A morning dip makes dealers SELL — the move accelerates.
- Price rips through your short strike.
- **The gate should have kept you FLAT today.**

GEX · THE EVIDENCE

Where GEX comes from

SqueezeMetrics (2016)

The original GEX white paper — introduced gamma exposure as a measurable market-structure signal and showed its link to realized volatility.

Barbon & Buraschi (2020)

'Gamma Fragility' — peer-reviewed evidence that dealer gamma positioning shapes intraday price stability and reversal behaviour.

◆ LIVE DEMO

Pull up Barchart's Gamma Exposure page for SPX. Show the net GEX value and the flip point. Ask: is net GEX positive (pinning regime) or negative (trending regime) right now? That single reading is the gate.

The cleanest fit — a native filter

✓ Fully native & sweepable

- OA has a built-in 'Net GEX >' filter
- Require net GEX above a threshold to sell
- 'Stand aside' = the gate simply doesn't fire
- Sweep the GEX threshold directly

⚙️ Settings to know

- GEX is computed off a strike window (default: 10 nearest)
- Confirm the window matches your intent
- **OA computes its OWN GEX — so your R2 open-interest question doesn't apply here**

Why this is Tier A:

PART 4

Putting It Together

Which edges cross over — and what you'll do tonight.

THE SCORECARD

Three edges, side by side

	Difficulty	Signal it needs	OA fit	Cross-over?
VRP	Foundational	IV minus realized vol	Tier B	Proxy only (IV Rank)
IV-Crush	Intermediate	High open IV + quiet day	Tier A/B	Yes, with slippage
GEX Switch	Advanced	Net dealer gamma	Tier A	Yes — native filter

The pattern: the harder the edge is to understand, the CLEANER it maps onto Options Alpha. GEX is advanced but the tool does the heavy lifting.

YOUR MANDATE

What you'll submit — and why

★ The signal-crossover test

An edge is “both-URL eligible” only if Options Alpha can compute its gating signal. GEX passes cleanly. IV-Crush passes via proxy. VRP passes weakly (structure only).

Do this first

Start with GEX

Cleanest OA validation. Build the Net GEX regime switch and sweep the threshold. Submit both URLs.

Do this second

Then IV-Crush

Reproduce your H1a filter with IV Rank + range proxy. Submit both URLs, note the proxy slippage.

R2-led

VRP on R2

The RV/IV signal lives on R2 / Options Omega. Validate structure on OA; own the edge on R2.

TONIGHT'S TAKEAWAY

Share the why, protect the what.

You now understand three real edges — why they pay, where the research backs them, and how to test them on Options Alpha. The structure is the craft; the signal is the moat.

01 **VRP**

Sell overpriced fear

02 **IV-Crush**

Pocket the deflation

03 **GEX**

Sell only when dealers pin